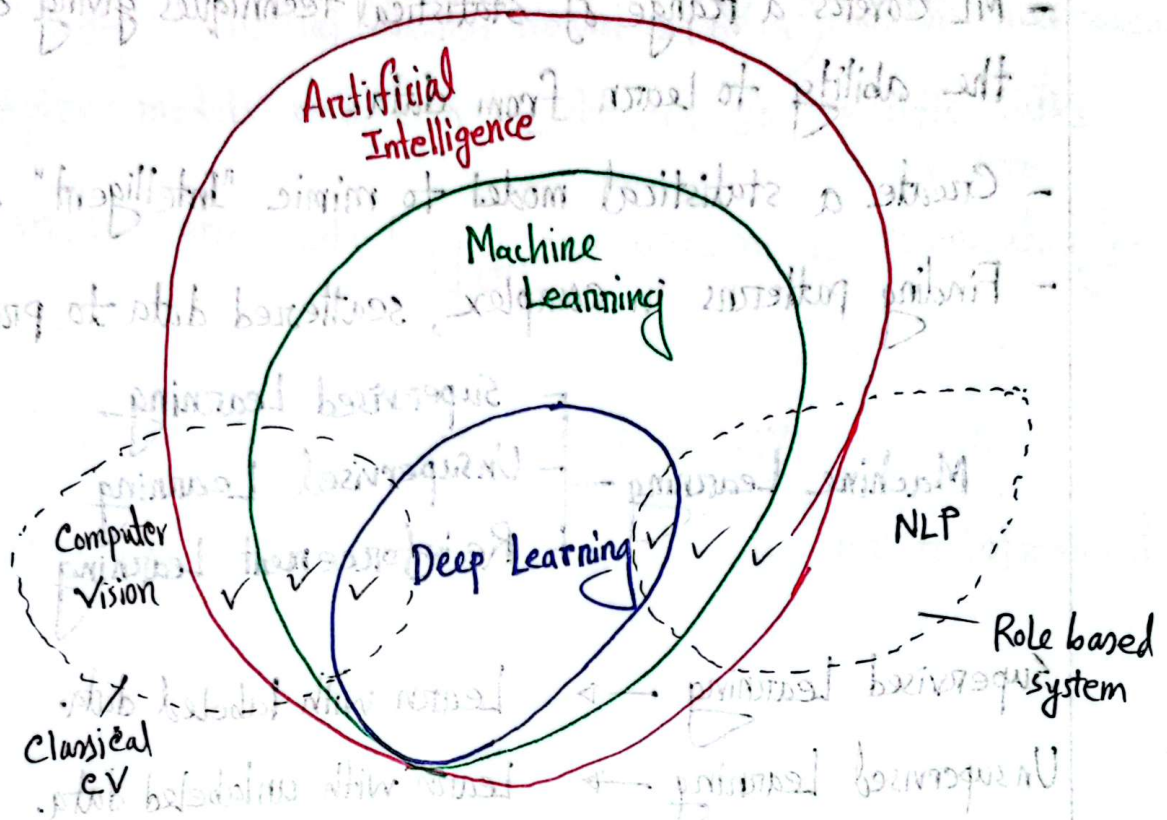


Introduction

Artificial Intelligence



Artificial Intelligence: Ability of a machine to imitate intelligent human behavior.

Machine Learning: Application of AI that allows a system to automatically learn and improves from experience.

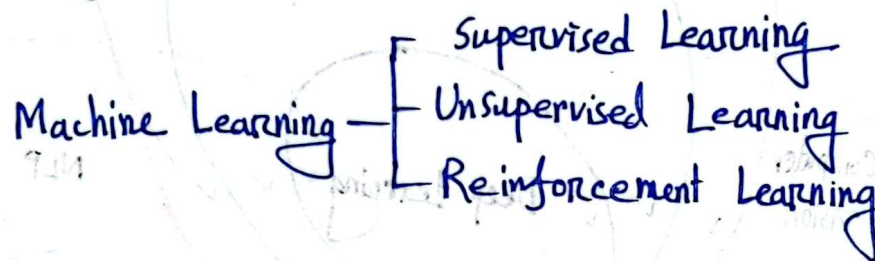
Deep Learning: Application of Machine Learning that uses complex algorithms and deep neural networks to train a model.

Sub-fields of AI

- | | |
|-------------------------|--------------------------------------|
| ① Machine Learning (ML) | ④ Natural Language Processing (NLP) |
| ② Deep Learning (DL) | ⑤ Automatic speech Recognition (ASR) |
| ③ Computer Vision (CV) | ⑥ Reinforcement Learning (RL) |

Machine Learning (ML)

- ML covers a range of statistical techniques giving computers the ability to learn from data.
- Create a statistical model to mimic "Intelligent" decision making.
- Finding patterns in complex, scattered data to present information



Supervised Learning → Learn with labeled data.

Unsupervised Learning → Learn with unlabeled data.

Reinforcement Learning → Learn through reward maximization.

ML Roadmap

1. Regression
2. Classification
3. Unsupervised Learning
4. Model evaluation
5. Feature Engineering
6. Hyperparameters Tuning
7. Neural Networks
8. Reinforcement Learning
9. NLP
10. Transformers

Deep Learning (DL)

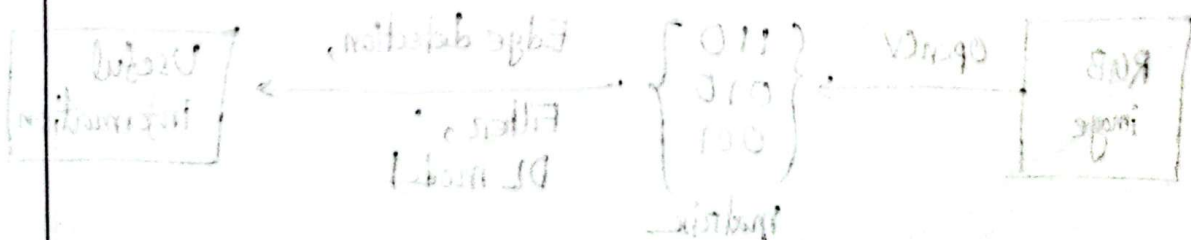
- DL using Artificial Neural Networks (ANNs) which are more complex models and can be trained for specific tasks.
- DL models are data hungry and are more computationally expensive.

Note: DL models are often used in CV and NLP.

→ DL models may supervised, unsupervised or reinforcement learning.

Some Application:

- Image Classifier.
- Image Colourisation.
- Automatic Machine Translation.



Computer Vision (CV)

→ Computer Vision is an interdisciplinary field concerning how computers can see and understand digital images and videos.

Example

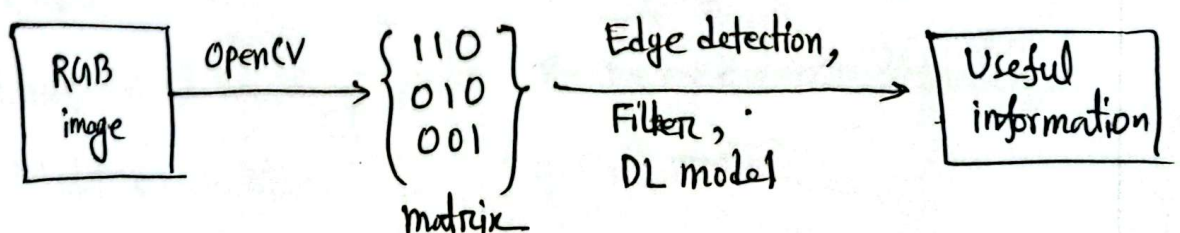
- Facial Recognition
- Image Segmentation
- Object Detection

Computer Vision $\left\{ \begin{array}{l} \text{Classical CV} \\ \text{CV using Deep Learning} \end{array} \right.$

** Computer Vision understands images & videos through -

Mathematical Algorithm + Deep Learning Models

- Popular CV library → OpenCV (open source)



Natural Language Processing (NLP)

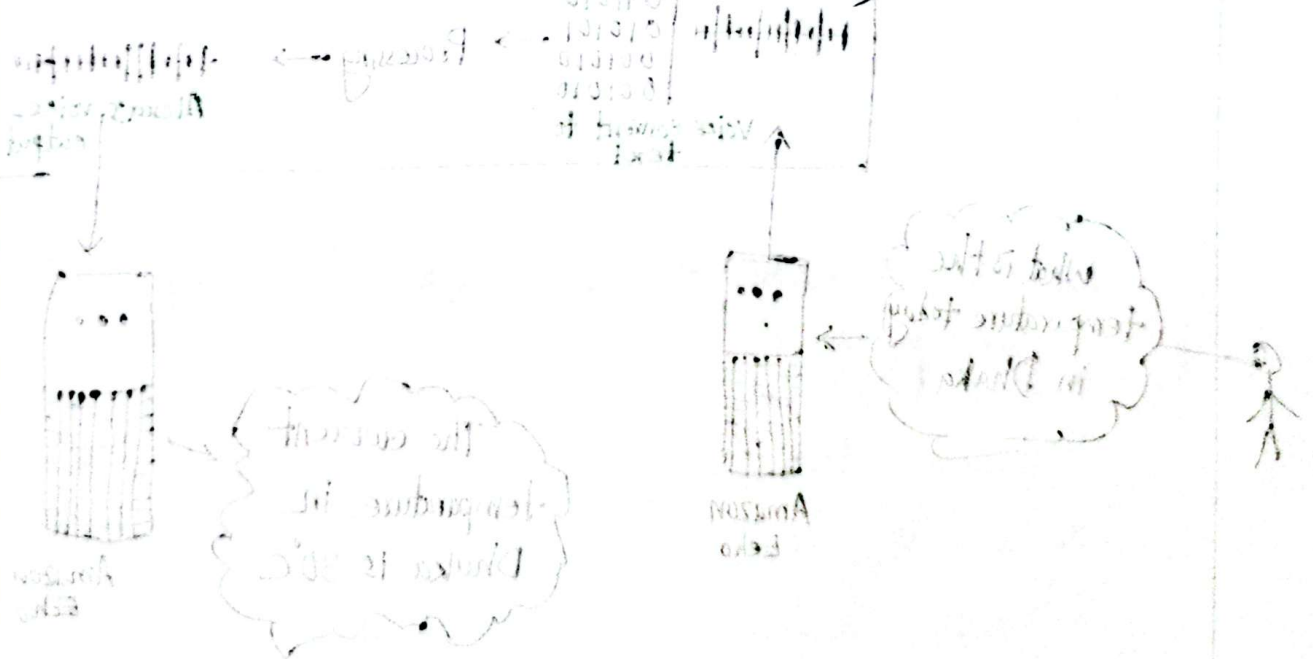
— NLP deals with the interaction between computers and human languages.

NLP separates into 2 sub-categories: —

1. Natural-language Understanding.
2. Natural language Generation.

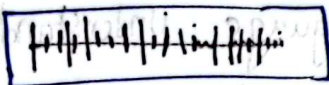
Ex:

- Named Entity Recognition (NER)
- Parts-of-Speech Tagging
- Machine Translation (Google translate)



Automatic Speech Recognition (ASR):

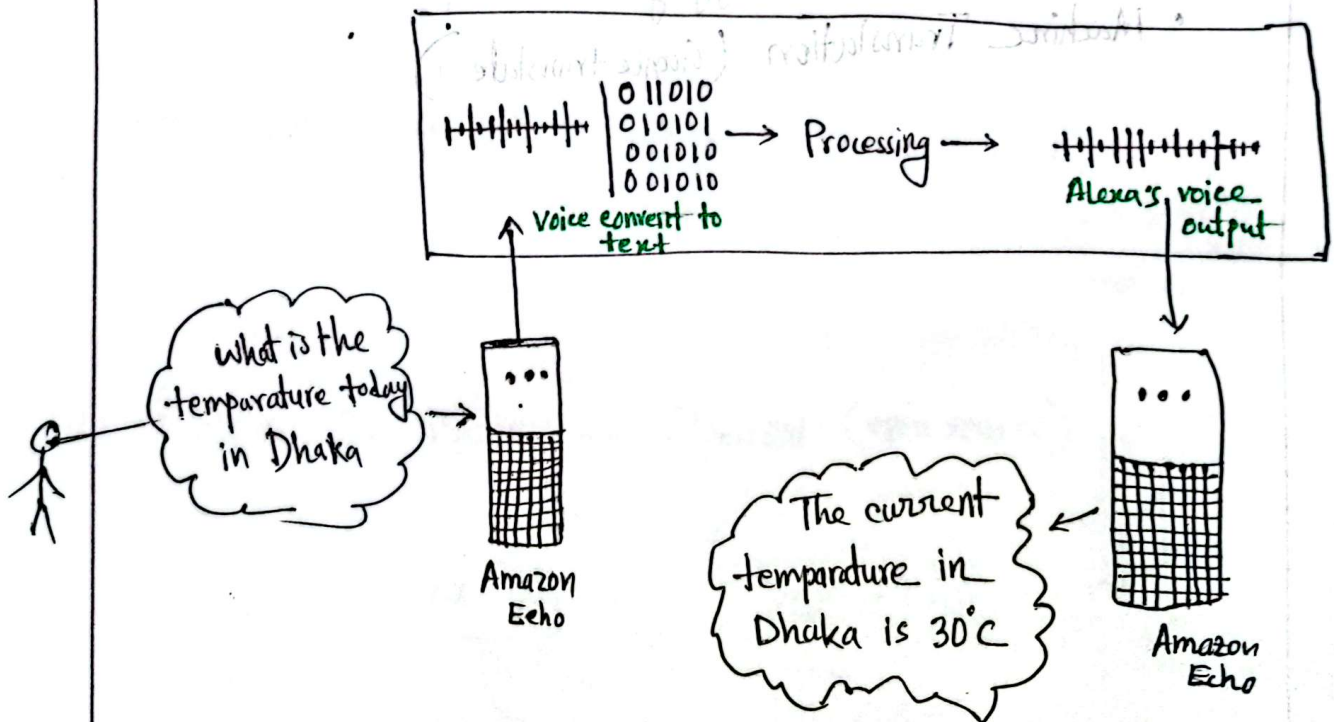
- ASR allows human beings to use their voice to speak with a computer interface in a way that, in its most sophisticated variations, resembles normal human conversation.

Voice →  → Text

Example

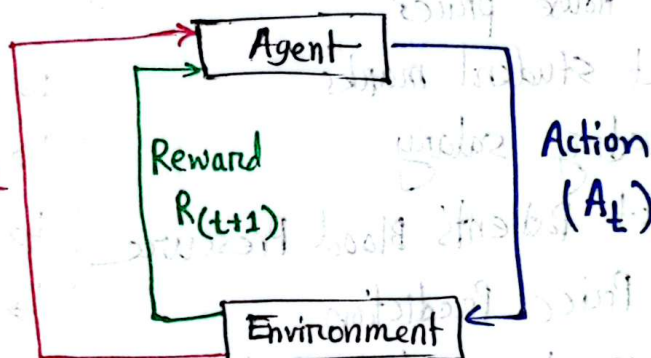
- Amazon Alexa

Q How Amazon Alexa works?



Reinforcement Learning (RL)

- The goal of RL is to train an agent to complete a task within an uncertain environment.
- The agent receives observations and a reward from the environment and sends actions to the environment.
- The reward measures how successful action is with respect to completing the task goal.



The goal of the Agent is - { Maximize the Reward over a long time .

Supervised & Unsupervised Learning

Supervised Learning:

— In supervised learning, we use labeled datasets to train algorithms that to classify data or predict outcomes accurately.

Regression and Classification are two task of supervised learning.

Regression: Predict a continuous numerical value.

- Ex:
- Predict House prices
 - Predict student marks
 - Estimating salary
 - Predict Patient's Blood Pressure
 - Stock Price Prediction
 - Weather temperature Prediction

Algorithm

- ↳ Linear Regression
- ↳ Polynomial Regression
- ↳ Decision Tree Regression
- ↳ Random Forest Regression
- ↳ Support Vector Regression (SVR)

Classification: Predict a discrete label or category.

- Ex:
- Spam Detection
 - Disease Prediction
 - Loan Approval
 - Sentiment Analysis
 - Animal Recognition (Cat, Dog, ...)
 - Traffic sign Recognition

Algorithm

- ↳ Logistic Regression
- ↳ Decision Trees
- ↳ Random Forest
- ↳ Support Vector Machine (SVM)
- ↳ Naive Bayes
- ↳ Neural Networks

Data $\left\{ \begin{array}{l} \text{Train Data} \rightarrow \text{it has labels} \rightarrow \text{So model can learn from these labels.} \\ \text{Test Data} \rightarrow \text{it doesn't have labels} \rightarrow \text{We try to predict labels.} \end{array} \right.$

Ex: Train data

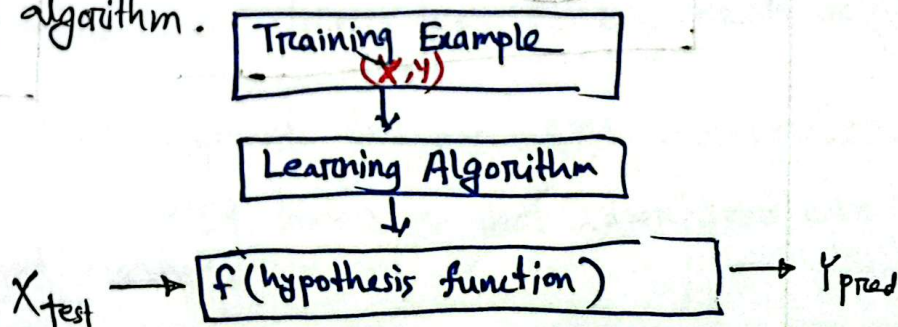
Time spent (x)	Marks (y)
5	16 $3x+1$
3	9.8 $3x+1 + \text{noise}$
4	11 $3x+1 + \text{noise}$
11	34 $3x+1$
7	22.5 $3x+1 + \text{noise}$

noise $\rightarrow$ Unmodelled features.
 noise $\rightarrow$ inaccurate measurements.

$$\therefore y = f(x)$$

$$\Rightarrow y = 3x+1 + \text{noise}$$

Note In supervised learning, the machine attempts to learn the relationships between x and y , by running labeled training data through a learning algorithm.



Unsupervised Learning:

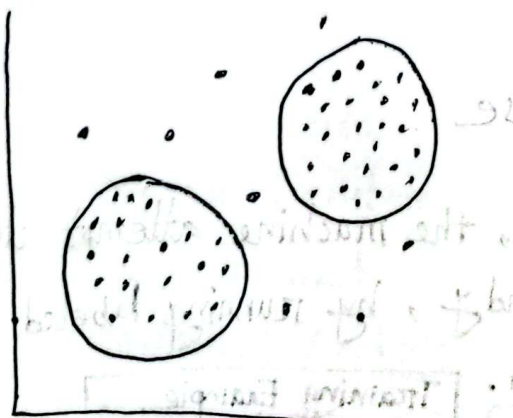
↳ The goal of unsupervised learning to find underlying ^{hidden} structure of the dataset, or how to we group the data most usefully starting with a unlabelled dataset.

Unsupervised learning can be broken down into 3 main task:-

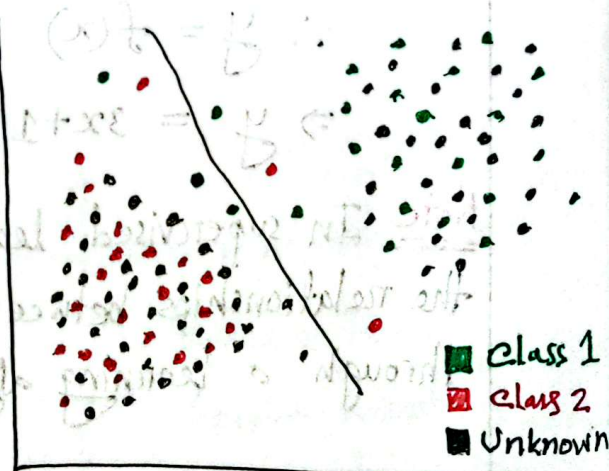
1. Clustering
 - K-means Clustering
 - Hierarchical clustering
2. Dimensionality Reduction
 - PCA
 - Singular Value decomposition
 - T-SNE
3. Association Rule Learning
 - Apriori
 - Eclat

Example

Unsupervised

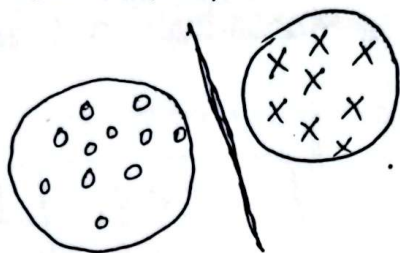


Supervised



Clustering :

↳ The goal of clustering is to create groups of data points such that points in different clusters are dissimilar while points within a cluster are similar.



Ex:

- Customer Segmentation
- Image Segmentation

Dimensionality Reduction

↳ It reduce the number of input features while preserving essential information.

Ex:

- Noise Reduction in images
- Preprocessing before supervised learning.

Association Rule Learning :

↳ Find interesting relationships/associations betⁿ variables in large dataset

- Ex:
- Market Basket Analysis (people who buy bread also buy butter)
 - Recommender systems.

Applications

- ① Google Photos wants to cluster similar looking faces under one person category.
- ① clustering news articles related to one topic under one heading.
- ① An advertising platform segments the population into smaller groups with similar purchasing habits so that advertisers can reach their target market.